Global Flood Impact Model Overview

CATBOOST GRADIENT BOOSTED REGRESSOR • EXECUTIVE SUMMARY & MODEL AUDIT

BEST TEST RMSE	BEST TRAINING RMSE	MODEL ARCHITECTURE
0.0985	**0.0695**	**CatBoost v1.2**
Iteration #1088 (Converged)	On full training history	1,200 Symmetric Trees

1. MODEL PURPOSE & OPERATIONAL SCOPE

The exported binary model (`catboost_flood_model.cbm`) is a specialized **Gradient Boosted Decision Tree (GBDT) regressor** trained on 5,503 global flood records. Its primary objective is to estimate the continuous **Flood Impact Index** using hydro-meteorological indicators, terrain dynamics, and regional climate attributes. The architecture uses symmetric trees with native categorical encodings, preventing data leakage across geographic borders.

2. EXTRACTED MODEL HYPERPARAMETERS

Hyperparameter	Configured Value	Operational Purpose
Tree Depth (`depth`)	7 Levels (Max 128 Leaves)	Captures complex interactions without deep overfitting.
Learning Rate (`learning_rate`)	0.03 (3.0%)	Ensures smooth gradient descent across 1,200 iterations.
L_2 Regularization (`l2_leaf_reg`)	5.0	Penalizes extreme leaf weights to prevent variance inflation.
Subsampling Rate (`subsample`)	0.80 (MVS Bootstrap)	Minimum Variance Sampling for robust tree bagging.
Categorical CTR Encoding	Borders & Counter (SkipTest)	Encodes climate zones and continent codes dynamically.

3. KEY PREDICTIVE DRIVERS

Ranked relative feature contributions derived from the model binary:

Feature Name	Importance
`Area (km²) / Log_Area`	
`Event_Duration_Days`	
`Cumulative_ET0_Evapotranspiration`	
`Total_Liquid_Rain_mm`	
`Estimated_Surface_Runoff_mm`	

4. VALIDATION & CONVERGENCE

- **Convergence Trajectory:** Initial test error started at `RMSE 1.0963` and decreased monotonically to `0.0985` at iteration #1088.
- **Overfitting Control:** The low gap between train RMSE (`0.0695`) and test RMSE (`0.0985`) confirms strong generalization on unseen test data.
- **Production Readiness:** Saved in static flatbuffer format (`cbm`), ready for zero-latency inference via C++, Python, or FastAPI microservices.

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